FP1 Chapter 3 Rational Functions and Curve Sketching

Lecture 3 Rational Functions and Curve Sketching

(Curve sketching, Asymptotes, Stationary Points, Related Functions, Inequalities)

Rational Function

- A rational function is defined as the quotient of two polynomials e.g $\frac{g(x)}{f(x)}$, where $g(x)$ and $f(x)$ are polynomials.

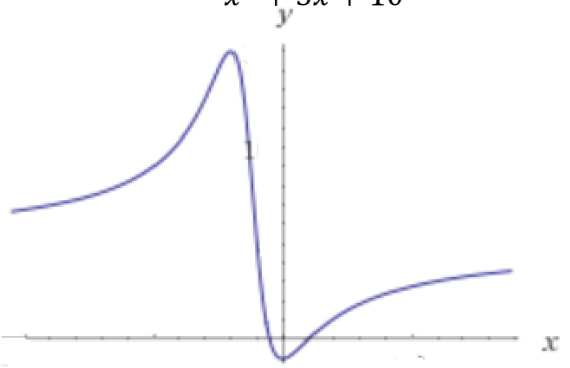
- $g(x)$ and $f(x)$ are polynomials of degree 0, 1 or 2. For examples,

$$\frac{A}{ax+b}, \frac{Ax+B}{ax+b}, \frac{A}{ax^2+bx+c}, \frac{Ax+B}{ax^2+bx+c}, \frac{Ax^2+Bx+C}{ax^2+bx+c}, \frac{Ax^2+Bx+C}{ax+b}$$

- Some graphs of rational function curves are as shown below:

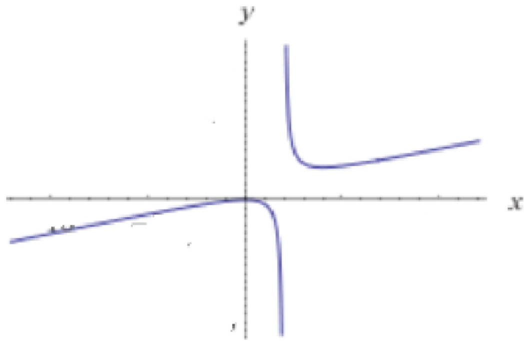
W2011 P13 Q10:

$$y = \frac{5(x^2-x-2)}{x^2+5x+10}$$



S2012 P13 Q6:

$$y = \frac{x^2}{x-2}$$



- In the examination, you are requested to sketch graphs, but the significant features of the graph (domain, range, intercepts, stationary points and asymptotes) must be shown.

Significant Features

(i) Domain

- All the possible values that x can take.

(ii) Range

- All the possible values that y can take.

- This can be done by rewriting the equation as a quadratic equation in x, we have $Ax^2 + Bx + C = 0$.

- In order that x be real, $B^2 - 4AC \geq 0$. Using this inequality we obtain the range of values of y.

(iii) Intercepts

- When $x = 0, y = ?$
- When $y = 0, x = ?$

(iv) Stationary points

- Stationary point when $\frac{dy}{dx} = 0$.
- Maximum point when $\frac{d^2y}{dx^2} < 0$.
- Minimum point when $\frac{d^2y}{dx^2} > 0$.

(v) Asymptotes

- Asymptotes is a line which becomes a tangent to a curve as x or y tends to infinity.
- As $x \to \pm\infty, y \to k$, k is a constant. Then $y = k$ is a **horizontal asymptote**.
- As $y \to \pm\infty, x \to k$, k is a constant. Then $x = k$ is a **vertical asymptote**.
- For example, $y = \frac{1}{x}$.

(A) Curve Sketching

1. Sketching Functions with $\boldsymbol{y = \frac{Ax+B}{ax+b}}$

- For **vertical asymptotes**, equate the denominator to zero, $x = -\frac{b}{a}$.
- For **horizontal asymptotes**, carry out long division.
- **Example 1:**

Sketch $y = \frac{2x+1}{2-x}$.

- **Example 2:**

Sketch $y = \frac{2-x}{2x+1}$.

- **Example 3:**

The curve C has equation $y = \frac{b(x+3a)}{x+a}$, where a and b are positive constants. State, in terms of a and b, the coordinates of the points where C intersects the axes, and the equations of asymptotes.

Sketch the curve C, showing its asymptotes.

2. Sketching Functions with $y = \frac{Ax^2+Bx+C}{ax^2+bx+c}$

- When the denominator is a quadratic expression,

  - There are **always two** vertical asymptotes.
  - The curve will normally cross the horizontal asymptotes.

- For **vertical asymptotes**, equate $ax^2 + bx + c = 0$. If $b^2 - 4ac < 0$, then there is no vertical asymptotes.

- For **horizontal asymptotes**, carry out long division.

- **Example 4:**

  Sketch $y = \frac{(x-1)(x-5)}{(x-2)(x-4)}$.

- **Example 5:**

If $y = \frac{x^2+1}{x^2-4}$ and x is real show that y cannot take values between $-\frac{1}{4}$ and 1. Sketch the graph of $y = \frac{x^2+1}{x^2-4}$ showing the asymptotes.

- **Example 6:**

The curve C has equation $y = \frac{a}{x} + \frac{b}{x^2}$, where a and b are constants, $a > 0, b \neq 0$.

(i) Show that C has exactly one stationary point and find its coordinates in terms of a and b.

(ii) On separate diagrams, draw a sketch of C for $b > 0$ and a sketch of C for $b < 0$, and in each case mark the coordinates of any intersections with the coordinate axes.

- **Example 7:**

The curve C has equation $y = \left(\frac{x+a}{x-a}\right)^2$.

(i) Show, by differentiation that C has exactly one stationary point and find the coordinates of this point.

(ii) Draw a sketch of C, showing its asymptotes.

- **Example 8:**

Given that $y = \frac{x^2+4}{(x-4)(x-1)}$. Find the stationary points and determine its nature of the stationary points.

Sketch the graph of $y = \frac{x^2+4}{(x-4)(x-1)}$, showing all the asymptotes.

- **Example 9:**

Given that $y = \frac{3x^2+10x+7}{x^2+2x+2}$. Find the stationary points and determine its nature of the stationary points.

Sketch the graph of $y = \frac{3x^2+10x+7}{x^2+2x+2}$, showing all the asymptotes.

Curves without Vertical Asymptotes

- Not all functions of the form $y = \frac{ax^2+bx+c}{px^2+qx+r}$ have vertical asymptotes.
- If the roots of $px^2 + qx + r = 0$ are not real, the curve will not have a vertical asymptote.
- **Example 10:**

Sketch the curve $y = \frac{2x^2+5x+3}{4x^2+5x+3}$, and find the range of possible values for y.

- **Example 10 (PYQ O/N 2017/11/12/13):**

9 The curve C has equation

$$y = \frac{3x-9}{(x-2)(x+1)}.$$

(i) Find the equations of the asymptotes of C. [2]

(ii) Show that there is no point on C for which $\frac{1}{3} < y < 3$. [4]

(iii) Find the coordinates of the turning points of C. [3]

(iv) Sketch C. [3]

3. Sketching Functions with $y = \frac{Ax^2+Bx+C}{ax+b}$

- Always has one vertical asymptote. E.g $ax + b = 0, x = -\frac{b}{a}$
- No horizontal asymptotes.
- There is an **oblique asymptote**. To find this asymptote, carry out long division. The quotient is the oblique asymptote.
- The shape of this curve must be one of the following cases.

- **Example 11:**

The curve C has equation $y = \frac{x^2+3}{x+1}$.

(i) Find the equations of the asymptote of C.

(ii) Find the values of x for which $\frac{dy}{dx} = 0$.

(iii) Draw a sketch of C, marking the coordinates of the turning points.

- **Example 12 (PYQ M/J 2016/11/12):**

7 A curve C has equation $y = \frac{x^2}{x-2}$. Find the equations of the asymptotes of C. [3]

Show that there are no points on C for which $0 < y < 8$. [4]

Sketch C, giving the coordinates of the turning points. [3]

- **Example 13 (PYQ O/N 2015/11/12/13):**

8 The curve C has equation $y = \dfrac{2x^2 + kx}{x + 1}$, where k is a constant. Find the set of values of k for which C has no stationary points. [5]

For the case $k = 4$, find the equations of the asymptotes of C and sketch C, indicating the coordinates of the points where C intersects the coordinate axes. [6]

(B) Graph of Related Functions

1. Graph of $\boldsymbol{y = |f(x)|}$

- From $y = f(x)$, reflect the curves for which $f(x) < 0$ in the x-axis, retain the section for which $f(x) > 0$.
- Example: $y = |(x + 1)(x - 2)(x - 5)|$

2. Graph of $y = f(|x|)$

- From $y = f(x)$, reflect the curves for which $x > 0$ in the y-axis, discard the section for which $x < 0$.

- Example: $y = (|x| - 1)(|x| - 3)$

- **Example 14:**

Sketch the curve whose equation is $y = f(x)$, where $f(x) = \frac{x+1}{x+2}$. Hence, sketch the curve whose equation is

(i) $y = |f(x)|$

(ii) $y = f(|x|)$

- **Example 15:**

Sketch the curve whose equation is $y = f(x)$, where $f(x) = \frac{3x-9}{(x-2)(x+1)}$. Hence, sketch the curve whose equation is

(i) $y = |f(x)|$

(ii) $y = f(|x|)$

- **Example 16:**

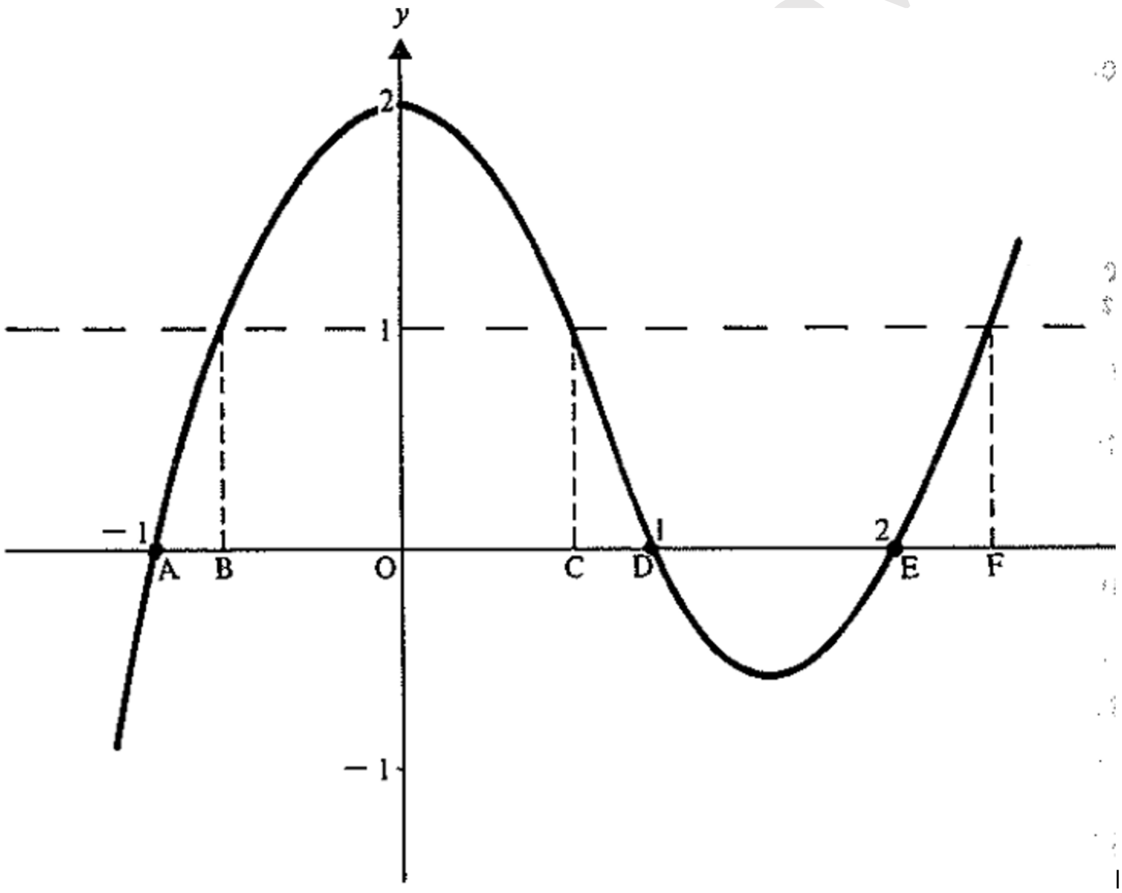
Sketch the curve whose equation is $y = f(x)$, where $f(x) = \frac{x^2}{x-2}$. Hence, sketch the curve whose equation is

(i) $y = |f(x)|$

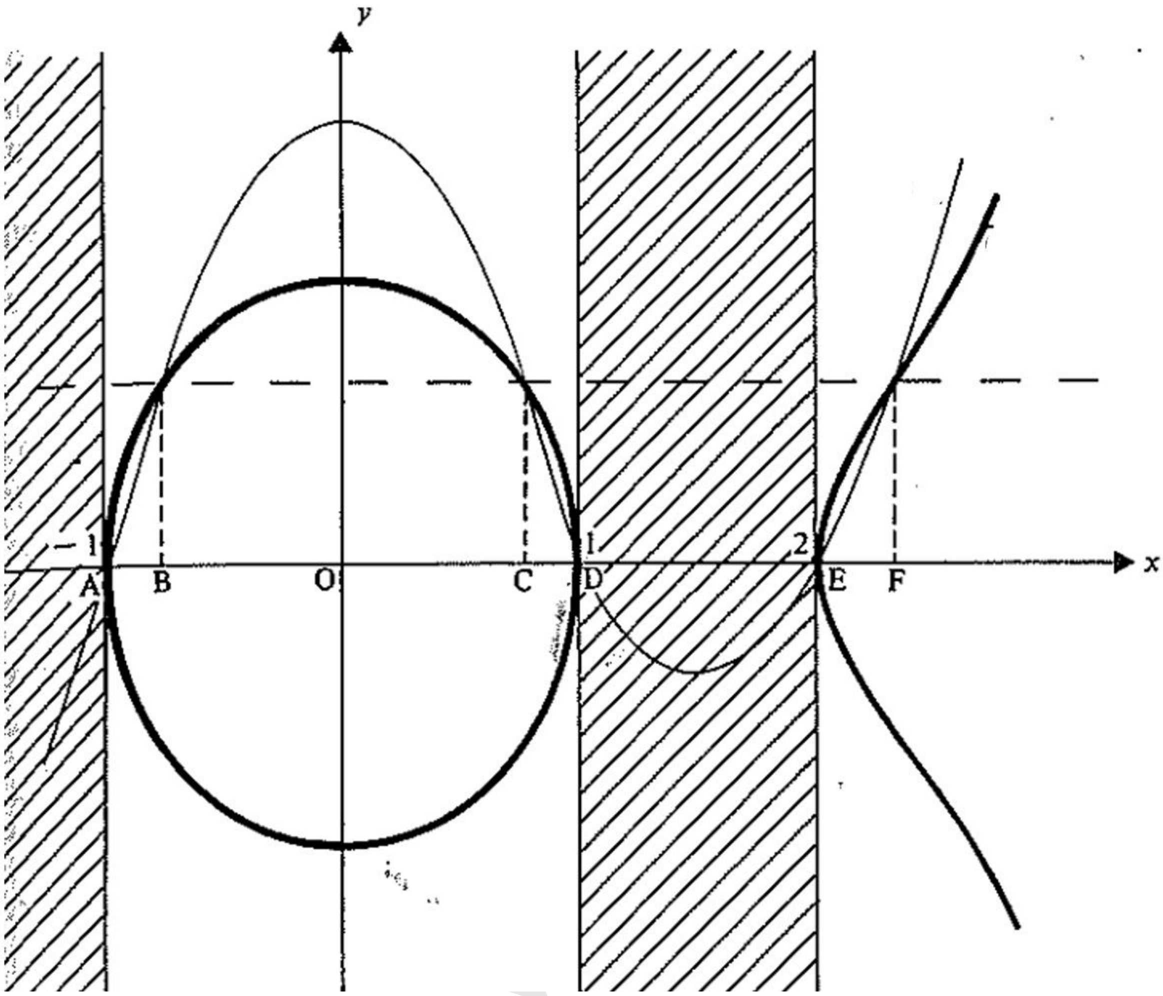
(ii) $y = f(|x|)$

3. Graph of $y^2 = f(x)$

- From $y^2 = f(x)$, since $y^2 \geq 0$, reflect the curves for which $y > 0$ in the x-axis, discard the section for which $y < 0$.
- The vertical asymptotes remain the same, but the equation of the horizontal asymptote $y = k$, where $k > 0$ will be changed to $y = \pm\sqrt{k}$.
- When the graph of $y = f(x)$ crosses the x-axis, the gradient of the graph is infinity. (The tangent is vertical.)
- Since $y^2 = f(x)$, then $y = \pm\sqrt{f(x)}$. The curve is symmetrical about the x-axis.
- When $f(x) > 1, \sqrt{f(x)} < f(x)$. But when $f(x) < 1, \sqrt{f(x)} > f(x)$.
- **Example**: $y^2 = (x+1)(x-1)(x-2)$.
- To start with $y^2 = f(x)$, sketch $y = (x+1)(x-1)(x-2)$ first as shown below.



- Note $f(x) \geq 0$, for $-1 \leq x \leq 1$ and $x \geq 2$, so domain of $y^2 = f(x)$ follows the same.
- Between A and B, C and D, E and F where $f(x) < 1$, so $\sqrt{f(x)} > f(x)$.
- Between B and C and right of F where $f(x) > 1$, so $\sqrt{f(x)} < f(x)$.
- The graph of $y^2 = (x+1)(x-1)(x-2)$ is shown below.



- Note that the graph of the curve $y = \sqrt{(x+1)(x-1)(x-2)}$ is the part of the curve which lies above the x-axis.

- **Example 17:**

Sketch the graph of

(i) $y = (x-1)(x-3)$

(ii) $y^2 = (x-1)(x-3)$

- **Example 18:**

Sketch the graph of

(i) $y = x^3 - 4x$

(ii) $y^2 = x^3 - 4x$

4. Graph of $y = \frac{1}{f(x)}$

- If (a, b) is a maximum point of the graph $y = f(x)$, where $b > 0$, then $\left(a, \sqrt{b}\right)$ is a minimum point of the graph $y = \frac{1}{f(x)}$.
- If (a, b) is a minimum point of the graph $y = f(x)$, where $b > 0$, then $\left(a, \sqrt{b}\right)$ is a maximum point of the graph $y = \frac{1}{f(x)}$.
- If the graph of $y = f(x)$ crosses the x-axis at the point $(a, 0)$, then $x = a$ is a vertical asymptote.
- Example: $y = \frac{1}{(x-1)(x-3)}$

- **Example 19:**

Sketch on separate diagrams the curves whose equations are

(i) $y = \frac{x}{1+x}$

(ii) $y^2 = \frac{x}{1+x}$

- **Example 20:**

Sketch the curve whose equation is $y = \frac{x}{x^2-1}$.

Hence, or otherwise, sketch the curve whose equation is $y = \frac{x^2-1}{x}$.

- **Example 21:**

Sketch the curve $y = \frac{x^2 - a^2}{x^2}$, where a is a positive constant. The equations of the asymptotes should be stated, together with the coordinates of any intersections with the axes.

Hence, sketch the curve $y^2 = \frac{x^2 - a^2}{x^2}$.

(C) Solving Equations and Inequalities Graphically

- In P1, you have learned how to solve simple quadratic inequalities such as

$$x^2 - 7x + 10 \geq 0.$$

- We established that we can add and subtract as usual with an inequality symbol, as if were an equal symbol. But to multiply or divide by a negative number, we must **change the sign of the inequality**.

- Hence, an inequality such as $\frac{ax+b}{cx+d} > 2$ cannot be solved simply by multiplying both sides of the inequality by $cx + d$, since we do not know whether

  (i) $cx + d$ is positive, giving $ax + b > 2(cx + d)$

  (ii) $cx + d$ is negative, giving $ax + b < 2(cx + d)$

- To solve inequalities such as $\frac{ax+b}{cx+d} > k$, we can use either of the following two methods:

  (i) Multiply both sides of the inequality by $(cx + d)^2$, which we know must be positive.

  (ii) Sketch $y = \frac{ax+b}{cx+d}$, solve $\frac{ax+b}{cx+d} = k$ and then, by comparing these two results, write down the solution to the inequality.

- **Example 22:**

A curve has equation $y = \frac{1}{x(x+2)}$.

(a) Write down the equations of all the asymptotes of C.

(b) The curve C has exactly one stationary point. The x-coordinate of the stationary point is -1.

(i) Find the y-coordinate of the stationary point.

(ii) Sketch the curve C.

Hence, solve the inequality $\frac{1}{x(x+2)} \leq \frac{1}{8}$.

- **Example 23:**

A curve has equation $y = \frac{x-1}{(x-2)(2x-1)}$. The line l has equation $y = \frac{1}{2}(x-1)$.

(a) Write down the equations of the asymptotes of C.

(b) By forming and solving a suitable cubic equation, find the x-coordinates of the points of intersections of l and C.

(c) Given that C has no stationary points, sketch C and l on the same axes.

Hence solve the inequality $\frac{x-1}{(x-2)(2x-1)} \geq \frac{1}{2}(x-1)$.

- **Example 24:**

If $y = \frac{x(x-3)}{x-4}$ and x is real show that y cannot take value between 1 and 9.

Sketch the curve $y = \frac{x(x-3)}{x-4}$.

On the same diagram, sketch the curve $y = \frac{5}{x}$ and find its intersection with the line $x = 4$.

Hence prove that the equation $\frac{5}{x} = \frac{x(x-3)}{x-4}$ has no positive root.

Exercises:

1. Find the coordinates of the turning points on the curve $y = \frac{4}{x+1} - \frac{1}{x-1}$.

   Write down the equations of the asymptotes of the curve.
   Sketch the curve.

2. Show that, for real values of x, $-\frac{1}{4} \leq \frac{x}{x^2+4} \leq \frac{1}{4}$.

   Sketch the curve $y = \frac{x}{x^2+4}$, showing the asymptotes and the coordinates of the turning points.

3. Prove that the curve with equation $y = \frac{(x+1)^2}{(x-1)(x+2)}$ has two points at which $\frac{dy}{dx} = 0$.

   Find the coordinates of these points and determine the nature of each point.
   Write down the equations of the asymptotes of the curve.
   Sketch the curve.

4. Write down the equations of the two asymptotes of the curve $y = x + 1 - \frac{4}{x-2}$, and find the coordinates of the points at which the curve meets the coordinate axes.
   Sketch the curve.

5. Find the constants A, B and C such that $\frac{x^2-9}{x-5} \equiv Ax + B + \frac{C}{x-5}$.

   Deduce, or find otherwise, the equations of the asymptotes of the curve $y = \frac{x^2-9}{x-5}$.
   Find the coordinates of the stationary points on this curve and sketch the curve.

6. Given that $y = \frac{x^2+3}{x-1}$, where x is real, show that y cannot take any real value between –2 and 6.

   Find the equations of the asymptotes of the curve $y = \frac{x^2+3}{x-1}$, and sketch the curve, showing the coordinates of the turning points.

7. The curve C has equation $y = 2x + 1 - \dfrac{5}{2x+1}$.

(i) Write down the equations of the asymptotes of C.

(ii) Show that $\dfrac{dy}{dx}$ is positive at all points of C.

(iii) Draw a sketch of C.

8. The curve C has equation $y = \dfrac{x}{(x+1)(x-2)}$. The line L has equation $y = -\frac{1}{2}$.

(a) Write down the equations of the asymptotes of C.

(b) The line L intersects the curve C at two points. Find the x-coordinates of these two points.

(c) Sketch C and L on the same axes.
(You are given that the curve C has no stationary points.)

Solve the inequality $\dfrac{x}{(x+1)(x-2)} \le -\dfrac{1}{2}$.

9. Given that the curve $y = \dfrac{4 - ax^2}{b + x}$ has asymptotes $x = -1$ and $y = 1 - x$, find the values of a and b.

Show that, at all points of the curve, $\dfrac{dy}{dx}$ is negative.

Sketch the curve.

10. The curve C has equation $y = \dfrac{x^2}{x^2 - a^2}$, where a is a positive constant.

(i) Show that C has one stationary point, whose coordinates are to be found.

(ii) Write down the equations of the asymptotes of C.

(iii) Show that, at all points of C, either $y > 1$ or $y \le 0$.

(iv) Draw a sketch of C.

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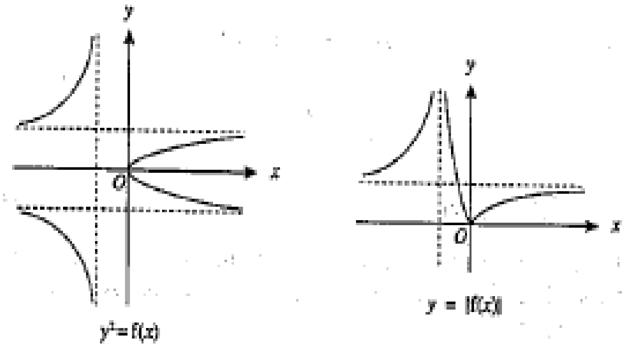

Fig. 2

The sketches in Fig. 2 show the graphs of $y^2 = f(x)$ and $y = |f(x)|$ for a certain function f.

(i) Sketch the graph of $y = f(x)$.

(ii) Sketch the graph of $y = f(|x|)$.

12. Show that, if $y = \dfrac{(x+1)(x-3)}{x(x-2)}$ and x is real, y cannot lie between 1 and 4. Sketch the curve $y = \dfrac{(x+1)(x-3)}{x(x-2)}$, showing the asymptotes and the coordinates of the turning points.

Hence sketch the curve $y = \left|\dfrac{(x+1)(x-3)}{x(x-2)}\right|$.

13. Sketch on separate diagrams the curves with equations

(i) $y = \dfrac{1}{x^2 - a^2}$,

(ii) $y^2 = \dfrac{1}{x^2 - a^2}$, where a is a positive constant.

You should state, in each case, the equation of any asymptote(s), and the coordinates of any stationary point(s).